pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Coefficients, P-Values, and Multicollinearity

WEEK 1 · DAY 3

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

How sure is each coefficient?

Every coefficient was fitted from 615 bonds. Another 615 would give other numbers.

Two readings that go too far

A small p-value: "this feature matters".

A large p-value: "this feature does nothing".

The recipe behind the spreads

In: rating, years to maturity, an issuer effect, noise.

Not in: sector. Not in: duration.

821 synthetic bonds of 150 invented issuers. Every coefficient today is an association in an invented file.

How statsmodels OLS works

Same line	The same least squares coefficients as scikit-learn
The extra	A standard error: how much a coefficient would change from one sample of bonds to another
Source	The spread of the residuals and how much each feature varies
You control	The features, and the constant column: sm.add_constant adds the intercept
What comes out	A results object: coefficients, standard errors, t, p-values, intervals

statsmodels: coefficients plus uncertainty

```
fit = sm.OLS(y_train, sm.add_constant(X_train)).fit()
fit.params, fit.bse, fit.pvalues
```

```
linear = LinearRegression().fit(X_train, y_train)
linear.intercept_, linear.coef_
```

12

coefficients

603

residual degrees of freedom

0.8847

R-squared, training rows

Without sm.add_constant: score 0.3055 instead of 0.2016, and no error.

Same least squares coefficients within 1e-9. statsmodels for inference, scikit-learn for pipelines.

How to read the coefficient table

t statistic	The coefficient divided by its standard error: standard errors from zero
p-value	How often a t this far from zero would turn up if the true coefficient were zero
Not said	A p-value says nothing about the size of the effect or its cause
Interval	About two standard errors either side: the coefficients the data does not rule out
What comes out	One row per coefficient: coef, std err, t, p-value, interval

Reading the coefficient table

Column	duration	Reads as
coef	0.0197	about 2 percent wider per year, other features fixed
std err	0.0033	typical move from sample to sample
t	5.96	standard errors from zero
P>|t|	0.000	below 0.0005, not zero
interval	0.0132 to 0.0262	values the data are consistent with

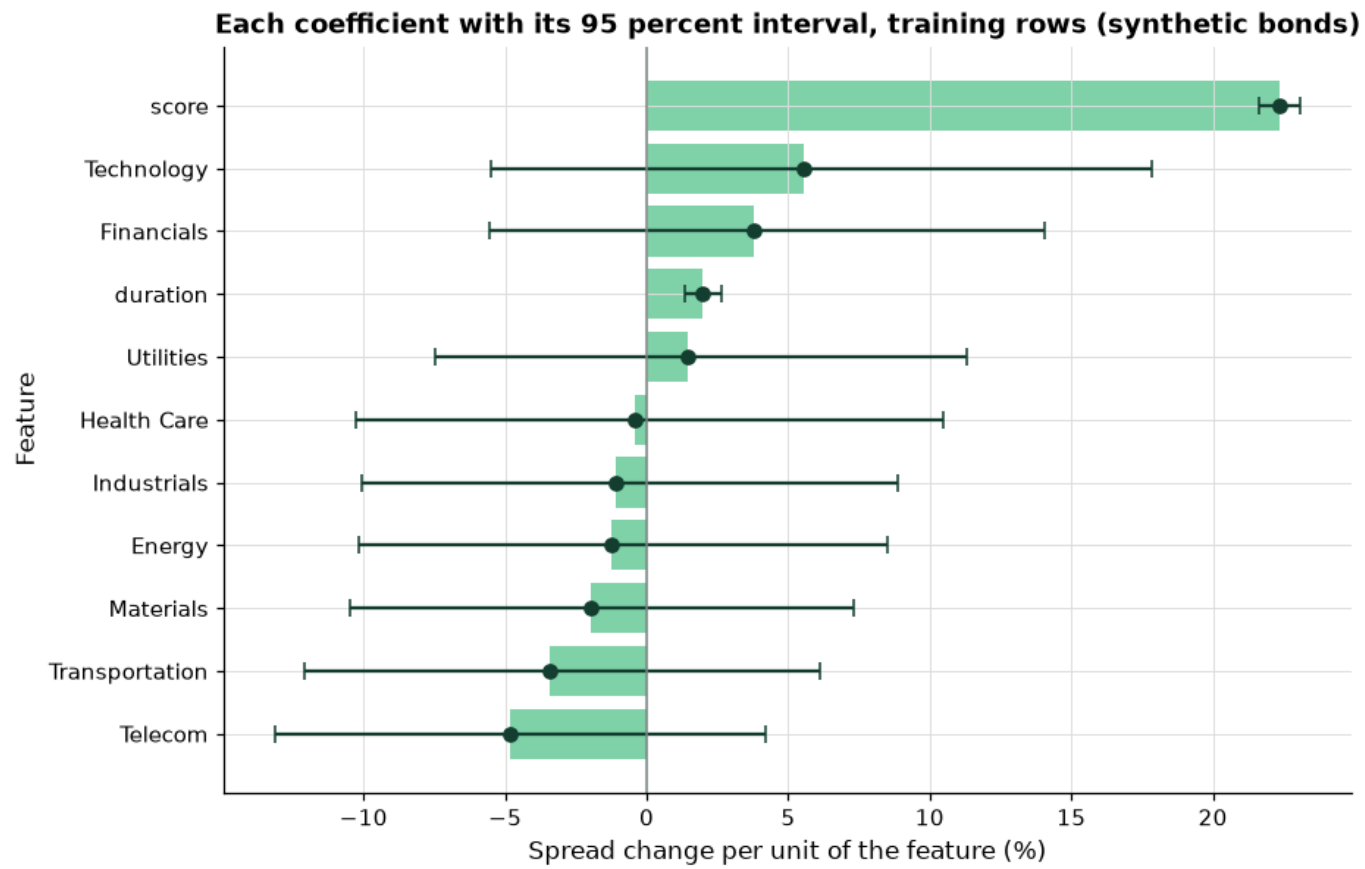
What a p-value says

If the true coefficient were zero and the assumptions held: the share of samples giving a t at least this far out.

What it is not

Not the probability the coefficient is real. Not a size. Not a cause. 0.05 is a convention.

Every coefficient with its interval



Clear of zero

score: 22.3 percent per notch. duration: 2.0 percent per year.

Across zero

All 9 sector dummies.

Rows have different units: compare each line with zero, not the bars with each other.

Sector: a feature the recipe never used

0 of 9

sector p-values below 0.05

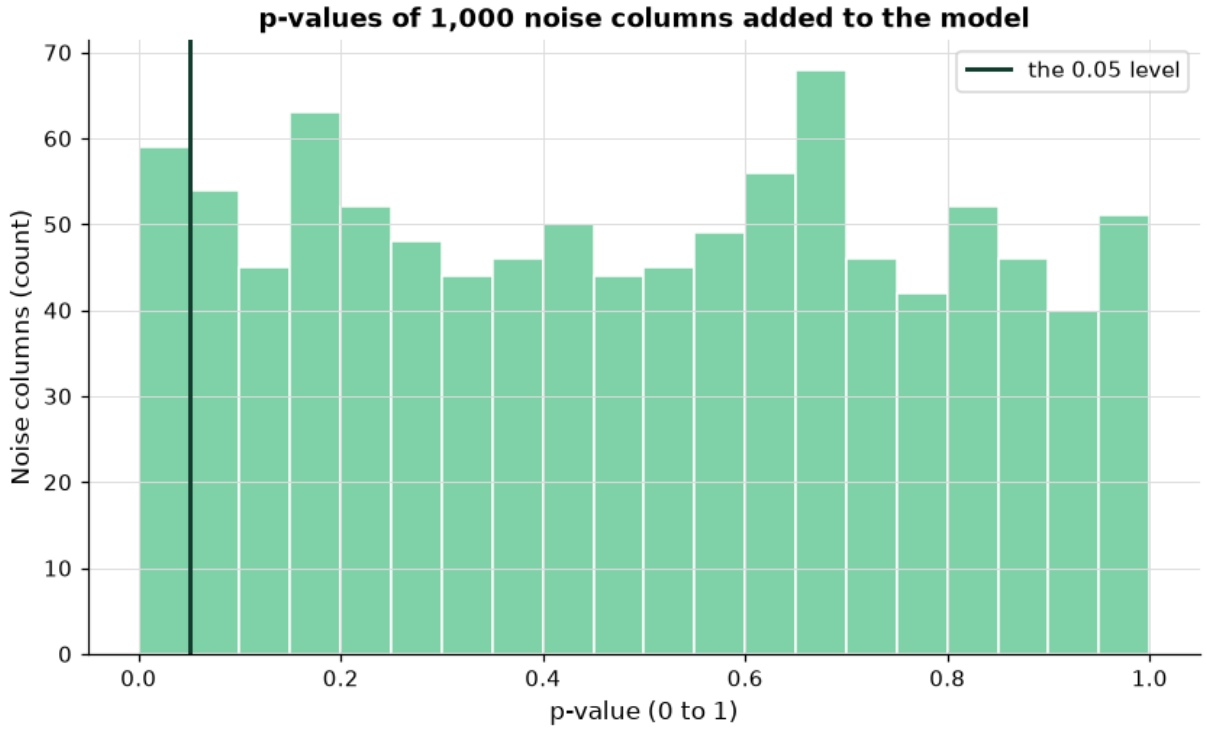
0.283

smallest p (Telecom)

Joint F-test, all nine together

F 0.82, p 0.599

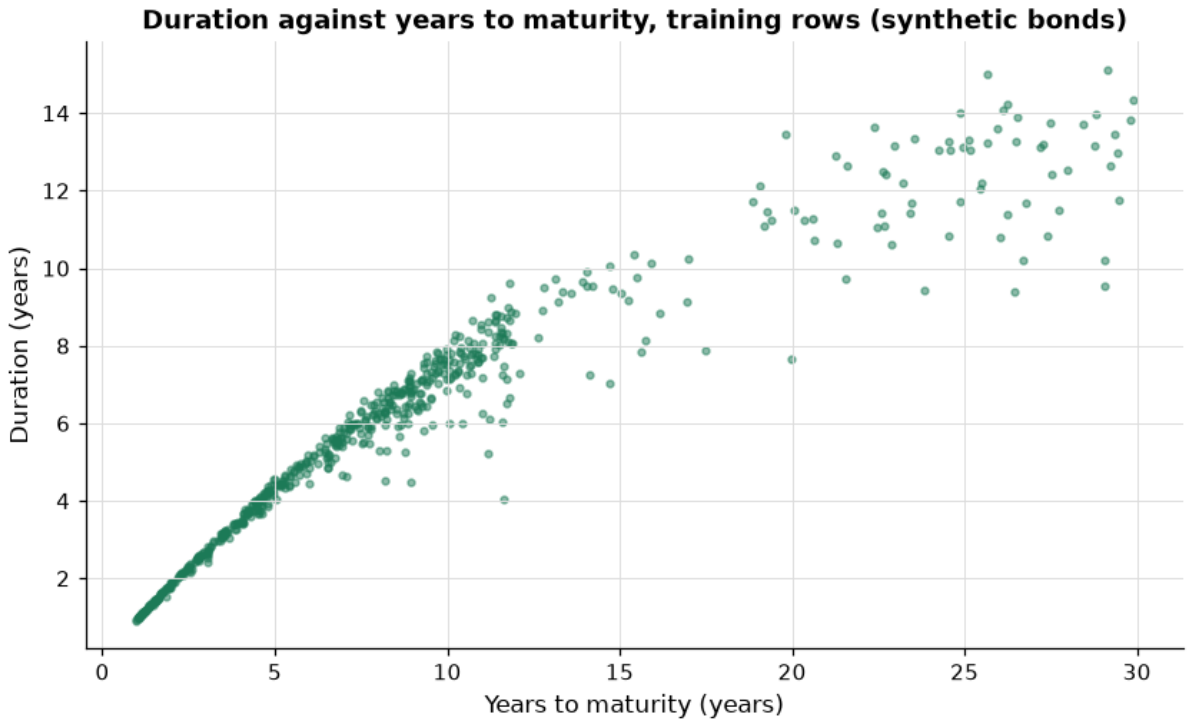
Largest sector p-value: 0.935. A large p does not prove zero.



59 of 1,000 noise columns came in below 0.05 (12 below 0.01): one in twenty, by chance.

Association is not cause

"Holding the other features fixed, a one-unit difference in x goes with a b difference in the fitted y, in this data."



0.9457

correlation of duration and years to maturity

Fair: bonds with longer duration have wider spreads in this file.

Not supported: duration widens the spread. The recipe uses maturity, not duration.

How multicollinearity is measured

The problem	Features move so closely together that the data cannot share their effect out
The cost	The line can still fit well, but each coefficient and its standard error become unstable
Correlation	A matrix of correlations finds the problem pair by pair
VIF	Regress one feature on the others: 1 / (1 - R-squared). It catches combinations of several
What comes out	A correlation per pair and a VIF per feature: above 5 high, above 10 very high

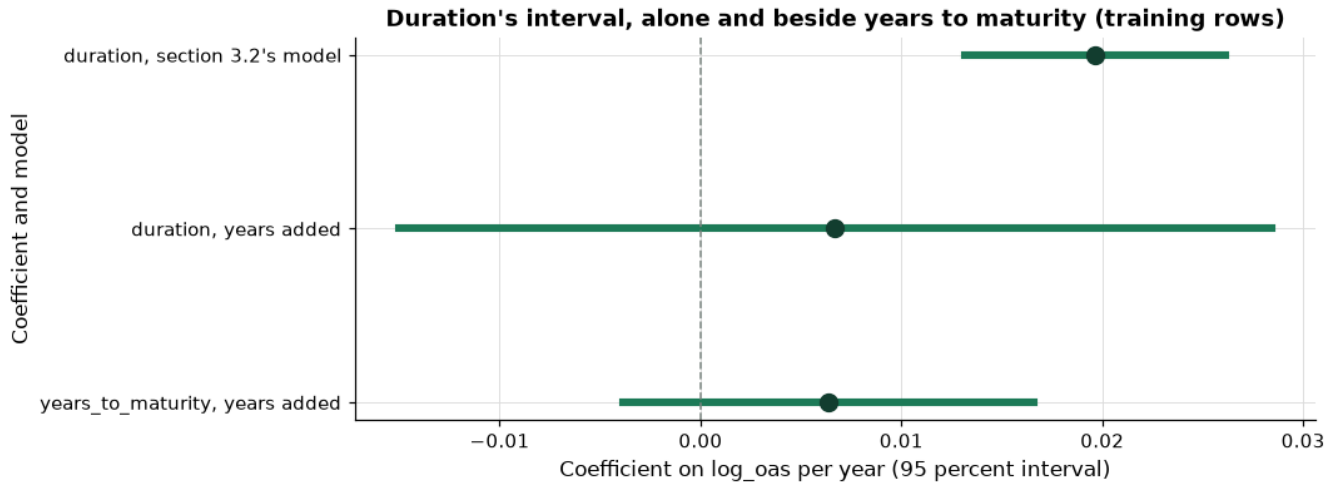
Multicollinearity: duration against years to maturity

Model	Feature	coef	std err	p
Duration only	duration	0.0197	0.0033	< 0.001
Both	duration	0.0067	0.0111	0.545
Both	years_to_maturity	0.0064	0.0052	0.22

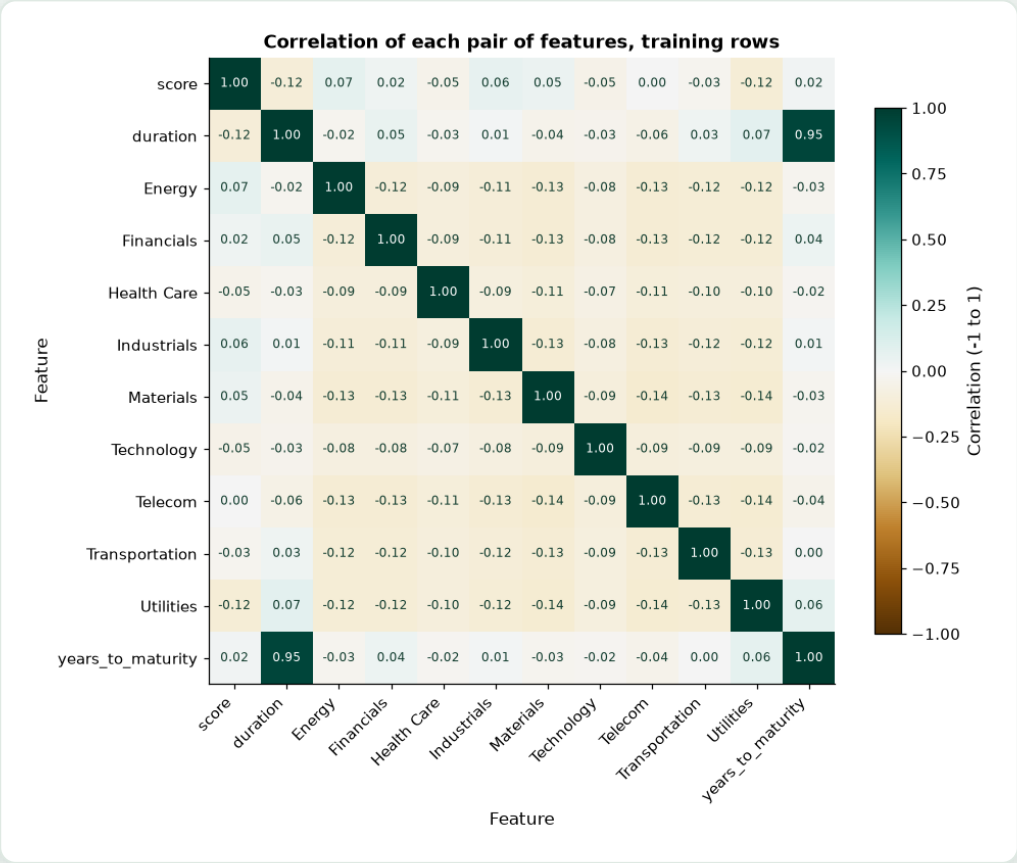
Both, tested together

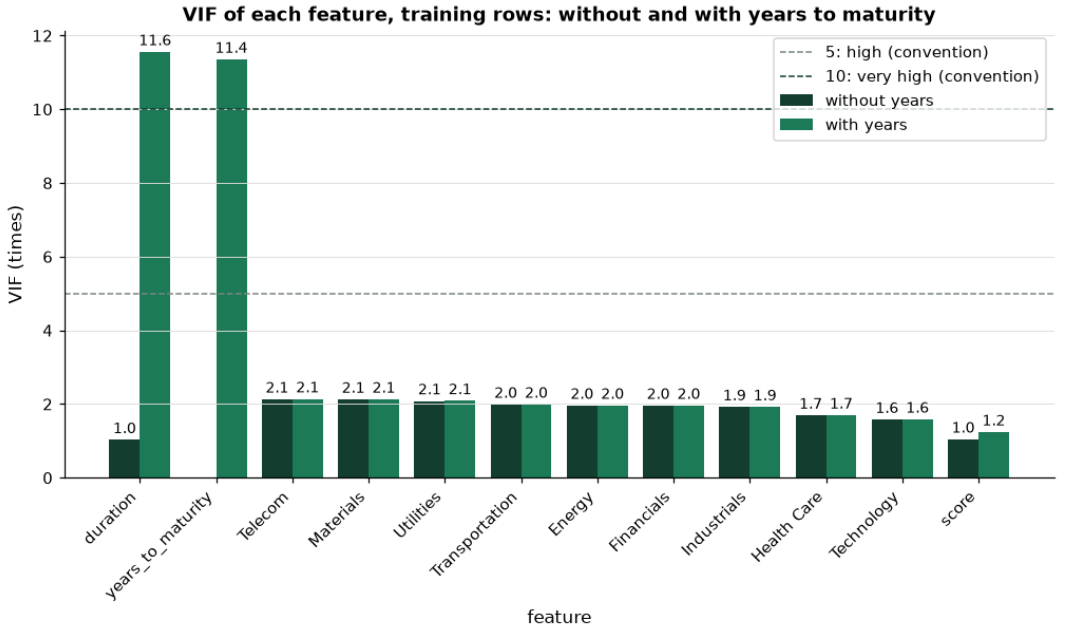
F 18.50

p < 0.001



Measuring the overlap: correlation and VIF





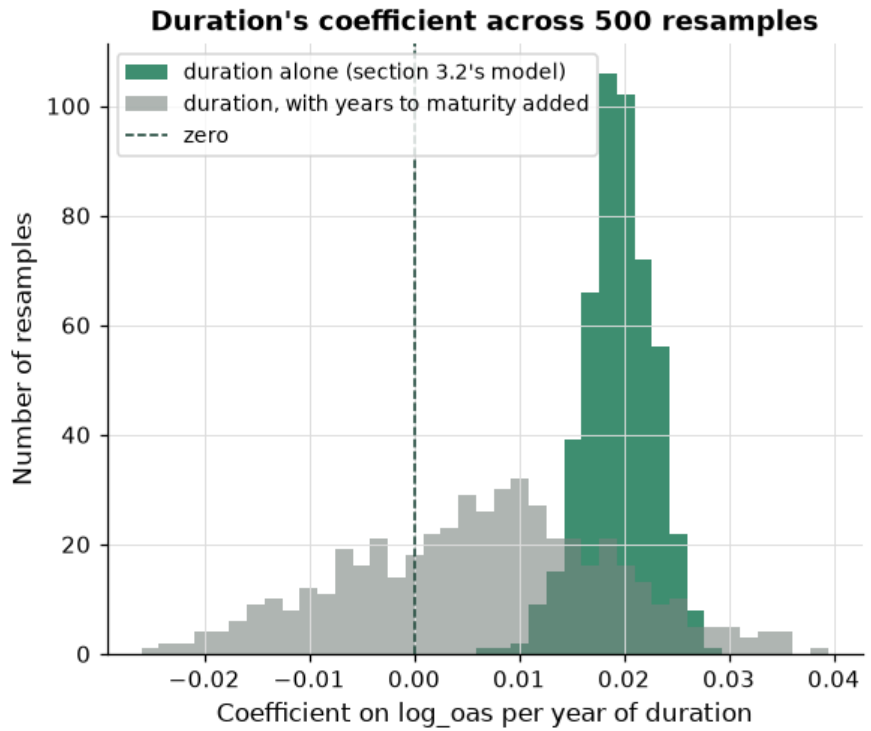

```
VIF_j = 1 / (1 - R2_j): j on all the others
```

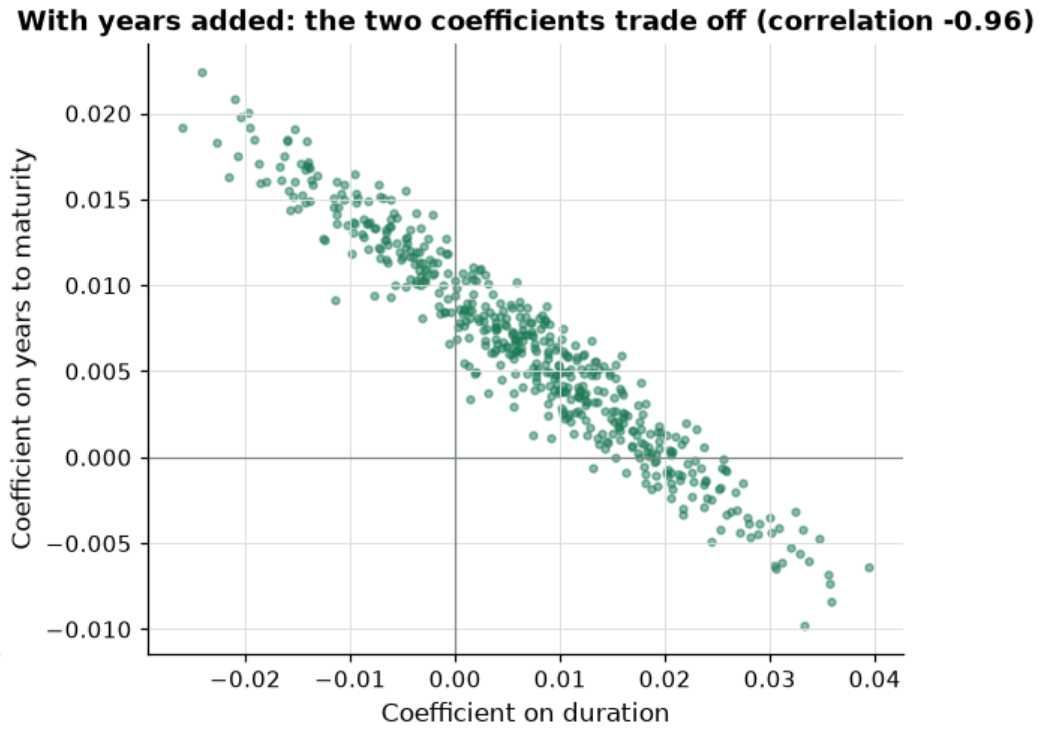
Above 5 high, above 10 very high. Duration's std err: 3.35x.

How the bootstrap works

The problem	A standard error describes other samples of bonds, and you have only one
Resample	Draw bonds from the training rows with replacement, to the same size
Refit	Fit the model on the new sample and record the coefficient; repeat 500 times
The reading	The spread of the 500 coefficients is a standard error you can see
You control	The number of resamples and the seed

Seeing a standard error: the bootstrap





The spread of 500 refits matches the standard error; with years added the pair trades off (correlation -0.96).

Perfect collinearity: spread_duration equals duration

spread_duration equals duration on 821 of 821 bonds: the VIF is 1 / 0.

statsmodels: warns, then splits in half

duration 0.0100 + spread_duration 0.0100 = 0.0201, the coefficient of duration alone.

SingularMatrixWarning: the parameters are not uniquely determined.

Excel LINEST: sets one aside (tested)

duration: coefficient 0, standard error 0. spread_duration: 0.0201.

Residual degrees of freedom 612, as if the copy were not there.

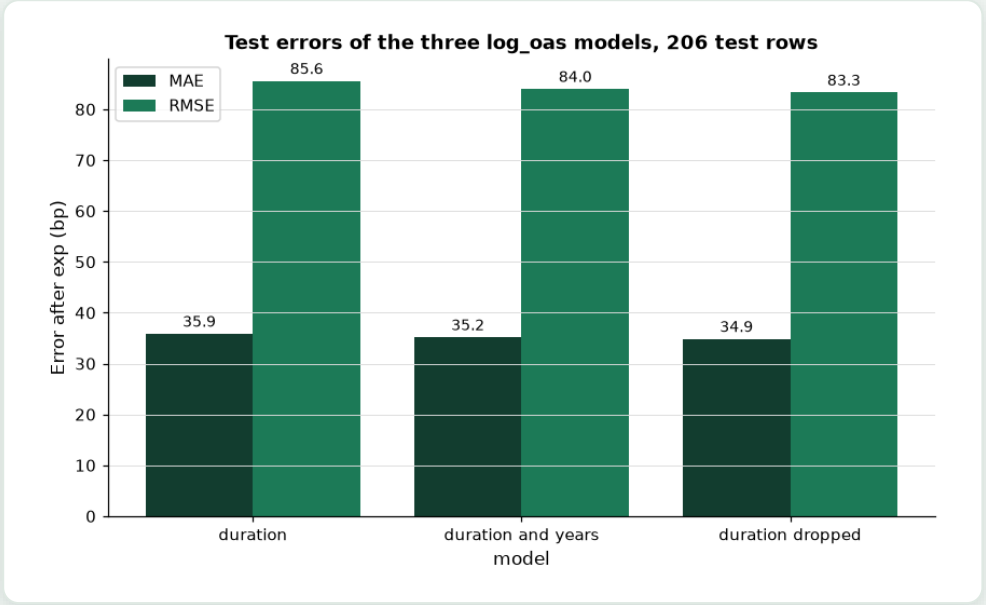
Different software, a different arbitrary answer, the same fitted values. Drop the copy.

Dropping a feature by VIF

Highest VIF on the training rows: duration (11.56) against years_to_maturity (11.35). Dropped: the recipe's own column is kept, by luck.

Model	R2 (log)	MAE (bp)	RMSE (bp)
duration	0.8690	35.9	85.6
duration and years	0.8690	35.2	84.0
duration dropped	0.8684	34.9	83.3

After the drop: highest VIF 2.13; years to maturity 0.0094 per year, standard error 0.0016.



Multicollinearity blurs the reading of coefficients, not the fitted values. The VIF cannot say which column is right.

Bonds of one issuer are not independent

46.0%

of the residual sum of squares shared within an issuer

24.2%

shuffled labels, on average (32.0% at most)

What that does to the table

OLS counts 615 independent bonds; for rating and sector there are about 149 issuers. Standard errors and p-values come out too small; the coefficients are not biased.

The standard fix, cluster-robust standard errors, is a later topic. Day 5 splits the rows by issuer.

Excel side by side

Excel side by side (tested in Excel)

`=LINEST(B2:B616,C2:M616,TRUE,TRUE)  (Q2)`	5 rows by 12 columns, reversed
`=T.DIST.2T(ABS(Q2/Q3),$R$5)  (Q9)`	duration p < 0.001
`=Q2-T.INV.2T(0.05,$R$5)*Q3`	duration from 0.0132
`=1/(1-INDEX(LINEST(D2:D616,HSTACK(C2:C616,E2:M616),...),3,1))`	VIF 1.03
`=((SSR without sectors - $R$6)/9)/($R$6/$R$5)`	F 0.82
`=F.DIST.RT(Q19,9,$R$5)`	p 0.599
`=LINEST(B2:B616,HSTACK(C2:D616,O2:O616),TRUE,TRUE)`	duration: 0, se 0

LINEST returns the columns in reverse: the last X column first, the intercept last.

Save the day: read what a commit changed

3. Experiment log: three log_oas models, with duration, with years to maturity added, and duration dropped

2. Add vif_table, with four tests: VIF, perfect collinearity, OLS against scikit-learn and Excel

1. Start of day 3: mlkit as it stands after day 2

```
git -C "{work}" log -p -1 -- mlkit.py
```

-p: the changes line by line. -1: one commit. -- mlkit.py: only commits that touched the module, so the experiment log is skipped.

14 passed at the end of the day

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: dts (spread duration x OAS) in the features: test R-squared 0.7193 against 0.5529, MAE 38.6 against 57.5 bp.

Code that runs is not the same as code that is right.

What you can do now

- statsmodels: scikit-learn's coefficients plus their uncertainty
- A p-value is not the chance the feature matters
- A large p-value does not prove zero
- Overlap blurs coefficients; the VIF measures it
- Bonds of one issuer: p-values too small

Week 1, Day 4 Regression assumptions and log spreads: the checks behind every p-value